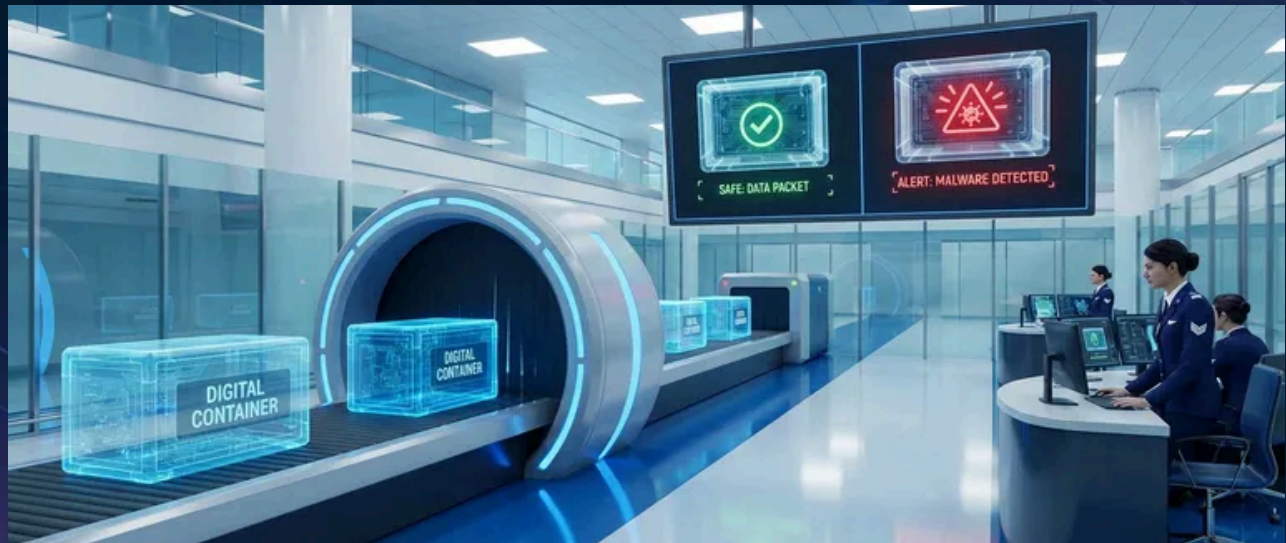


Shift-Left Security: Container Scanning in CI



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Webstack
Builders

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The appeal of shift-left security is obvious: finding vulnerabilities early in development costs less to fix than finding them in production. Container scanning belongs in your CI pipeline, not as an afterthought in staging.

The practical challenge is equally obvious. Teams enable container scanning and immediately face 47 critical vulnerabilities. Half are in base image packages the application never uses. A quarter have no available fix. The rest require major version bumps that would take weeks to properly test. The team disables scanning “temporarily” and never re-enables it.

This pattern repeats constantly. The scanner becomes noise that developers ignore or bypass. Security metrics look terrible because nobody’s actually scanning. The tool that was supposed to improve security has made it worse by creating learned helplessness.

WARNING

The fix isn’t better scanners – it’s better policies that lead to actionable findings. A scanner that reports 200 unfixable CVEs teaches developers to ignore all security alerts. A scanner that reports 3 fixable criticals with clear remediation steps gets those vulnerabilities fixed the same day.

This article covers how to integrate container scanning that developers will actually use: scanner selection, policy configuration that balances security and velocity, exception workflows for the inevitable unfixable CVEs, and metrics that tell you whether your security posture is actually improving.

Understanding Container Vulnerabilities

Where Vulnerabilities Hide

Container images are layered, and vulnerabilities can exist at every layer. Understanding where they come from helps you prioritize what to fix and what to accept.

Here’s the uncomfortable reality for a typical Node.js application: about 60% of vulnerabilities are in layers developers don’t directly control.



Layer	% of CVEs	Fixable by Developer
Application code	5%	Yes
npm dependencies	35%	Partially (transitive deps are tricky)
Node.js runtime	10%	Base image choice
OS packages (apt)	40%	Base image choice
Base image (Debian)	10%	Base image choice

Vulnerability distribution in a typical Node.js container.

Your application code – the part you write and fully control – accounts for roughly 5% of CVEs. The npm packages you directly depend on add another 35%, but many of those are transitive dependencies you’ve never heard of. The remaining 60% comes from the OS packages and base image, which most developers treat as a black box.

This distribution explains why “just fix all the CVEs” is impractical. You can’t fix a vulnerability in Debian’s libssl by editing your code. You need to update your base image, hope the distro has patched it, or accept the risk.

CVE Scoring and Severity

CVSS (Common Vulnerability Scoring System) scores tell you how bad a vulnerability **could** be in the worst case. They don’t tell you how bad it is **for you**.

Severity	CVSS Range	Typical Example
Critical	9.0 - 10.0	Remote code execution, no auth required
High	7.0 - 8.9	Privilege escalation, auth required



Severity	CVSS Range	Typical Example
Medium	4.0 - 6.9	Information disclosure, specific conditions
Low	0.0 - 3.9	Minor info leak, local access required

CVSS severity levels.

The problem is context. A CVSS (Common Vulnerability Scoring System) 9.8 critical in an XML parser doesn't matter if your application only parses JSON. A CVSS (Common Vulnerability Scoring System) 6.5 medium in your authentication library matters a lot if you're running a public-facing API.

CVSS Score	NVD Severity	Your Priority	Reason
9.8	Critical	Maybe High	Is the vulnerable function called?
9.5	Critical	Low	Package not used at runtime
7.2	High	Critical	Actively exploited in the wild
6.5	Medium	High	Exposed on public internet
4.0	Medium	None	Internal tool, no network access

CVSS scores vs contextual priority.

INFO

CVSS (Common Vulnerability Scoring System) measures theoretical severity; your priority should consider exploitability, exposure, and whether the vulnerable code path is reachable in your application.



Some CVEs have been rated critical for years with no known exploits. Others get weaponized within days of disclosure. The CVSS (Common Vulnerability Scoring System) score alone doesn't capture this – you need to check whether there's an active exploit in the wild (CISA's Known Exploited Vulnerabilities catalog <<https://www.cisa.gov/known-exploited-vulnerabilities-catalog>> is useful here) and whether the vulnerability applies to your deployment context.

Types of Findings

Scanners detect different categories of issues, and each requires a different response.

➤ OS package vulnerabilities

These vulnerabilities come from apt, apk, or yum packages in your base image. These are the bulk of findings and often the hardest to fix directly. Your remediation is usually updating the base image or switching to a slimmer alternative.

➤ Language dependency vulnerabilities

These vulnerabilities come from npm, pip, go modules, or Maven packages. These are more actionable – you can often fix them by updating a version in your manifest file. The complication is transitive dependencies: a vulnerability in ``nth-check`` that you've never heard of, pulled in by ``css-select``, which is pulled in by ``cheerio``, which you actually use.

➤ Configuration issues

These vulnerabilities are Dockerfile problems: running as root, including secrets in the image, using ``latest`` tags. These are fully within your control and should be fixed immediately.

➤ Secrets and credentials

These vulnerabilities are hardcoded passwords, API keys, or certificates baked into the image. These should fail builds unconditionally – there's no legitimate exception for shipping credentials in container images.

➤ Malware and supply chain attacks

These vulnerabilities are rare but catastrophic. The ``event-stream`` incident showed how a compromised maintainer can inject malicious code into popular packages. Scanners increasingly check for known malicious packages.

Scanner Selection

Popular Scanner Comparison

The container scanning landscape has consolidated around a few tools. Here's how they compare:



Trivy (Aqua Security)

has become the de facto open-source choice. It's fast, ships as a single binary, and scans OS packages, language dependencies, secrets, and IaC configurations. The breadth of coverage and zero-config startup make it ideal for CI pipelines. The main downside is noise – without tuning, you'll see a lot of findings.



Grype (Anchore)

is focused and fast. It pairs with Syft for SBOM generation and has good accuracy. It's less feature-rich than Trivy but might be preferable if you want a scanner that does one thing well.



Clair (Red Hat/Quay)

is battle-tested in the Quay registry but shows its age. Setup is more complex than newer alternatives, and it's slower. Unless you're already invested in the Red Hat ecosystem, there's little reason to choose it for new projects.



On the commercial side, **Snyk** offers excellent developer experience with fix PRs, prioritization, and IDE integration. **Aqua** (the company behind Trivy) offers runtime protection, policy engines, and enterprise features. Both come with enterprise pricing.

Scanner	Speed	Coverage	CI Integration	Cost
Trivy	Fast	OS, Lang, Secrets, IaC	Excellent	Free
Grype	Fast	OS, Lang	Good	Free
Clair	Moderate	OS	Moderate	Free
Snyk	Moderate	OS, Lang, Code	Excellent	Paid
Aqua	Moderate	Full stack	Excellent	Paid

Scanner comparison by key criteria.



✓ SUCCESS

Trivy has become the de facto open-source choice for CI/CD scanning due to its speed, breadth of coverage, and zero-config startup. Start there unless you have specific requirements.

Running Trivy in CI

Trivy's GitHub Action makes integration straightforward. Here's a pattern that works well: run the scan twice. The first pass reports all findings to GitHub's Security tab without failing the build. The second pass is the actual gate, configured to fail only on fixable critical vulnerabilities. For GitLab CI, the pattern is similar.

```
1  name: Security Scan
2
3  on:
4    push:
5      branches: [main]
6    pull_request:
7
8  jobs:
9    trivy-scan:
10     runs-on: ubuntu-latest
11     steps:
12       - uses: actions/checkout@v4
13
14       - name: Build image
15         run: docker build -t myapp:${{ github.sha }} .
16
17       - name: Run Trivy vulnerability scanner
18         uses: aquasecurity/trivy-action@master
19         with:
20           image-ref: 'myapp:${{ github.sha }}'
21           format: 'sarif'
22           output: 'trivy-results.sarif'
23           severity: 'CRITICAL,HIGH'
24           exit-code: '0' # Report only, don't fail
25
26       - name: Upload Trivy scan results
```




```
27     uses: github/codeql-action/upload-sarif@v3
28     with:
29       sarif_file: 'trivy-results.sarif'
30
31   - name: Trivy gate check
32     uses: aquasecurity/trivy-action@master
33     with:
34       image-ref: 'myapp:${{ github.sha }}'
35       severity: 'CRITICAL'
36       ignore-unfixed: true
37       exit-code: '1' # Fail on fixable criticals
```

The `ignore-unfixed: true` (GH Action) option and `--ignore-unfixed` flag (GitLab CI) are crucial. Without it, you'll fail builds on CVEs that have no available patch – frustrating developers with problems they can't solve.

Policy Configuration

Defining Security Gates

The scanner itself is just a tool. The policy – what you do with the findings – determines whether scanning improves security or becomes noise.

Trivy supports a configuration file that centralizes policy decisions:

.trivy.yaml

```
1  severity:
2    - CRITICAL
3    - HIGH
4
5  # Don't fail on unfixable vulnerabilities
6  ignore-unfixed: true
7
8  # Skip directories that don't ship to production
9  skip-dirs:
10    - test/
11    - docs/
12
```



```
13  # Custom ignore file for acknowledged risks
14  ignorefile: .trivyignore
15
16  timeout: 10m
17  format: table
18  exit-code: 1
```

Trivy configuration file for CI.

The ignore file is where you document exceptions. Every entry should explain **why** it's ignored:

```
# CVE in test dependency, not shipped to production
CVE-2023-12345

# Disputed CVE, vendor confirms not exploitable in our context
CVE-2023-67890

# No fix available, risk accepted until Q2 2024
# Ticket: SEC-123
# Expires: 2024-06-30
CVE-2023-11111

# False positive - we don't use the affected function
CVE-2023-22222
```

Trivy ignore file with documented exceptions.

WARNING

Every ignore entry should have a comment explaining why it's ignored and a ticket tracking the exception. Undocumented ignores become permanent blind spots.



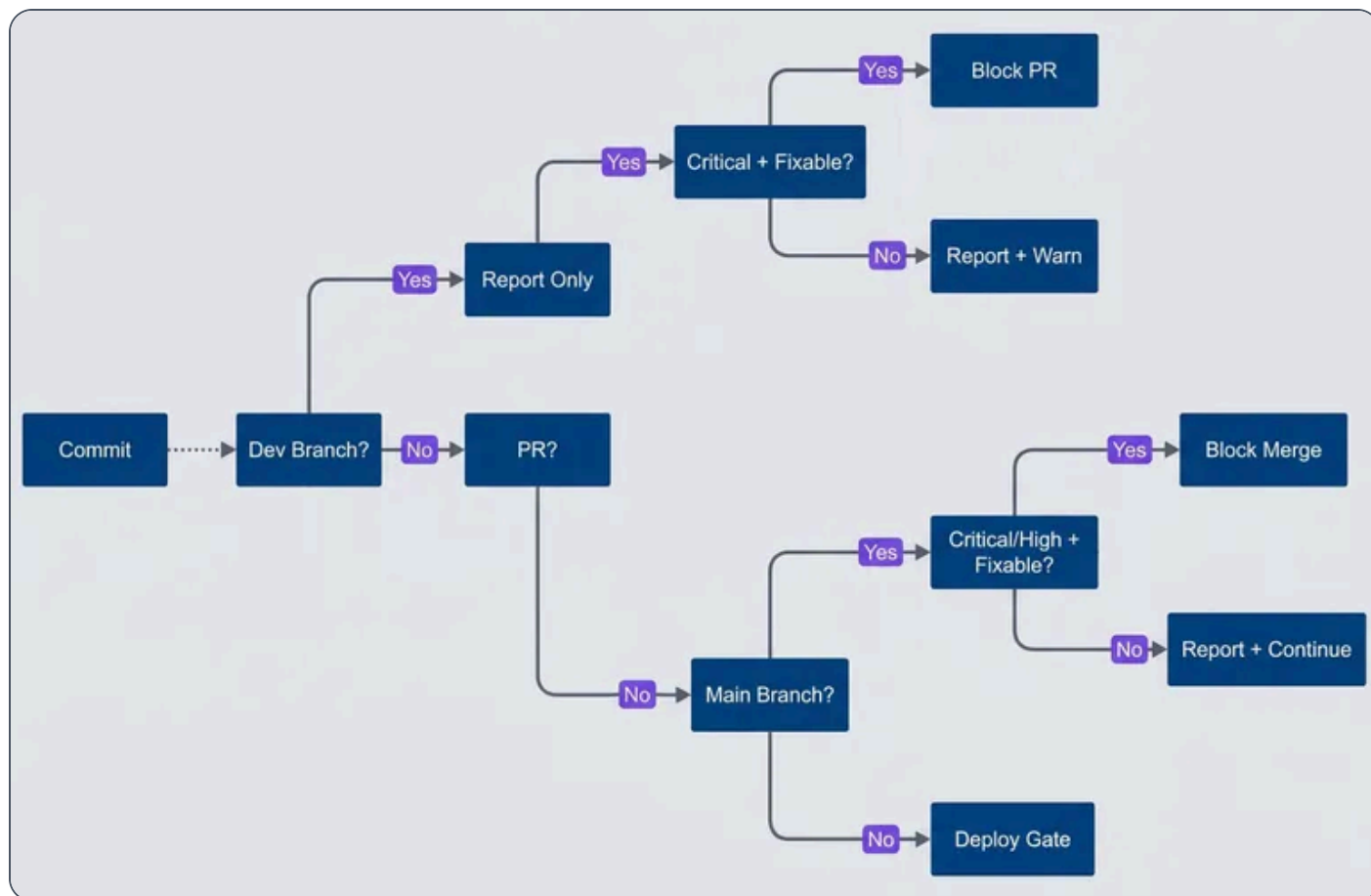
Tiered Severity Gates

Not every pipeline stage needs the same policy. Development branches should be permissive – you don't want to block a developer from iterating on a feature because of a CVE (Common Vulnerabilities and Exposures) in the base image. Production deployments should be strict.

Here's a tiered approach that balances security and velocity:

- **Development branches:**
Report only, never fail. Developers see findings but aren't blocked. This builds awareness without creating friction.
- **Pull requests:**
Fail on fixable critical vulnerabilities only. This catches new problems the developer introduced without blocking them on pre-existing issues.
- **Main branch:**
Fail on fixable critical and high vulnerabilities. This is the gate to production – stricter than PRs but still ignoring unfixable issues.
- **Production deployment:**
Strictest gate. Same as main branch, plus SBOM generation and checks against known exploit databases.





Tiered security gate decision flow.

The key insight is that you're not lowering security by being permissive on dev branches – you're increasing adoption. A scanner that developers trust and use beats a strict scanner that gets disabled.

Exception Workflows

Some vulnerabilities can't be fixed immediately. Maybe there's no patch. Maybe the fix requires a breaking change that needs a full release cycle. You need a process for handling these legitimately rather than pretending they don't exist.

The workflow should have four steps:



1

Developer request:

The developer adds the CVE to `.trivyignore`` with a comment explaining why, a ticket tracking the exception, and an expiration date.

2

Security review:

Someone on the security team reviews the request. Is the vulnerable code path reachable? Is there a workaround? What's the actual risk in your context? Is a fix expected soon?

3

Approval and tracking:

If approved, the exception is documented in a central location with calendar reminders for the expiration date.

4

Periodic review:

Monthly review of all active exceptions. Is a fix now available? Has the risk profile changed? Should we extend or close the exception?

The exception reasons I see most often:

Not exploitable:	The vulnerable code path isn't reachable in your application
No fix available:	Waiting for upstream to release a patch
False positive:	Scanner incorrectly identified the vulnerability
Mitigated:	Other controls prevent exploitation (WAF rules, network segmentation)
Risk accepted:	Business decision to accept the risk with compensating controls



 INFO

Exceptions should expire. A CVE (Common Vulnerabilities and Exposures) ignored six months ago might have a fix now, or the risk landscape might have changed. Default to 90-day expiration with required re-review.

SBOM Generation

What Is an SBOM

A Software Bill of Materials is an ingredients list for your software. It answers: what packages are included, what versions, what licenses, and where did they come from?

Three scenarios make SBOMs worth the effort:

1

Incident response:

When log4shell dropped, teams scrambled to figure out which services were affected. With SBOMs, you can answer "are we running log4j?" by searching a database instead of scanning every image in your registry. The difference between a 30-minute response and a 3-day audit.

2

Compliance:

Enterprise customers ask for SBOMs in security questionnaires. License compliance (GPL, Apache, MIT) requires knowing what's in your software. Export control regulations may apply to certain cryptographic libraries. Having SBOMs ready turns these from multi-week projects into quick lookups.

3

Supply chain security:

SBOMs let you track the provenance of components and detect unexpected changes. If a package suddenly appears that wasn't there before, you want to know about it.

Generating SBOMs with Trivy

Trivy can generate SBOMs in multiple formats:



sbom-commands.sh

```
1  # SPDX format (common for license compliance)
2  trivy image --format spdx-json --output sbom.spdx.json myapp:latest
3
4  # CycloneDX format (common for security)
5  trivy image --format cyclonedx --output sbom.cdx.json myapp:latest
6
7  # Generate from filesystem without building image
8  trivy fs --format cyclonedx --output sbom.cdx.json .
9
10 # Scan an existing SBOM for vulnerabilities
11 trivy sbom sbom.cdx.json
```

SBOM (Software Bill of Materials) generation and scanning commands.

The last command is particularly useful. Instead of scanning images repeatedly, you can scan the SBOM (Software Bill of Materials) – which is much faster since it's just a JSON file. When a new CVE (Common Vulnerabilities and Exposures) drops, scan all your SBOMs to find affected services in seconds.

For CI integration, generate the SBOM (Software Bill of Materials) alongside your image and attach it as an attestation using cosign:

.gitlab-ci.yml

```
1  sbom:
2    stage: security
3    script:
4      - trivy image
5        --format cyclonedx
6        --output sbom-${CI_COMMIT_SHA}.json
7        $CI_REGISTRY_IMAGE:$CI_COMMIT_SHA
8
9    # Attach SBOM to container image as attestation
10   - cosign attest
11     --predicate sbom-${CI_COMMIT_SHA}.json
12     --type cyclonedx
13     $CI_REGISTRY_IMAGE:$CI_COMMIT_SHA
14
15  artifacts:
```



```
16     paths:
17     - sbom-${CI_COMMIT_SHA}.json
```

CI pipeline generating SBOM (Software Bill of Materials) attestation.

INFO

Cosign (from the Sigstore project) cryptographically signs your SBOM (Software Bill of Materials) and attaches it to your container image in your OCI (Open Container Initiative) registry. It uses keyless signing via your CI provider’s OIDC identity, so there are no keys to manage. The signed attestation (stored with a `.att` extension) proves the SBOM (Software Bill of Materials) came from your pipeline and hasn’t been tampered with.

SBOM Format	Use Case	Tool Support
SPDX	License compliance, legal	Broad
CycloneDX	Security focus, VEX	Growing
Syft JSON	Anchore ecosystem	Anchore tools

SBOM format comparison.

For most teams, CycloneDX is the right choice. It’s designed with security in mind and supports VEX (Vulnerability Exploitability eXchange) documents for communicating which vulnerabilities actually affect your deployment. VEX lets you publish statements like “CVE (Common Vulnerabilities and Exposures)-2023-12345 does not affect us because we don’t use the vulnerable XML parsing feature”—turning your `.trivyignore` decisions into a machine-readable format that downstream consumers can use.



Base Image Strategy

Choosing Secure Base Images

The single biggest factor in your CVE (Common Vulnerabilities and Exposures) count is your base image choice. A full Ubuntu image ships with hundreds of packages you'll never use – and each one is a potential vulnerability.

base-image-comparison.dockerfile

```
1  # X BAD: Full OS image - hundreds of packages
2  FROM ubuntu:22.04
3  # CVEs: ~50-100 typically
4  # Size: ~77MB compressed
5
6  # ⚠ BETTER: Slim variant - fewer packages
7  FROM python:3.11-slim
8  # CVEs: ~20-50 typically
9  # Size: ~45MB compressed
10
11 # ✅ GOOD: Alpine - minimal packages
12 FROM python:3.11-alpine
13 # CVEs: ~5-15 typically
14 # Size: ~17MB compressed
15 # Caveat: musl libc compatibility issues
16
17 # ✅ BEST: Distroless - no shell, minimal
18 FROM gcr.io/distroless/python3
19 # CVEs: ~0-5 typically
20 # Size: ~15MB compressed
21 # Caveat: No shell for debugging
22
23 # ✅ ALTERNATIVE: Chainguard images
24 FROM cgr.dev/chainguard/python:latest
25 # CVEs: Usually 0
26 # Size: ~25MB compressed
27 # Caveat: Free for :latest only; pinned versions require subscription
```

Base image options by security profile.



✓ SUCCESS

Distroless images contain only your application and its runtime dependencies – no shell, no package manager, no unnecessary utilities. This dramatically reduces attack surface and CVE (Common Vulnerabilities and Exposures) count.

Multi-Stage Builds for Security

Multi-stage builds let you use a full toolchain for building while shipping a minimal image to production. The build stage has compilers, package managers, and dev dependencies. The production stage has only what's needed to run.

Dockerfile

```
1  # Stage 1: Build with full toolchain
2  FROM node:20 AS builder
3  WORKDIR /app
4  COPY package*.json ./
5  RUN npm ci
6  COPY . .
7  RUN npm run build
8
9  # Stage 2: Production with minimal image
10 FROM gcr.io/distroless/nodejs20-debian12
11 WORKDIR /app
12 COPY --from=builder /app/dist ./dist
13 COPY --from=builder /app/node_modules ./node_modules
14 COPY --from=builder /app/package.json ./
15
16 CMD ["dist/index.js"]
```

Multi-stage build separating build-time and runtime dependencies.

The vulnerability reduction is dramatic. A single-stage build using `node:20` might have 60 CVEs: 45 in OS packages, 3 in Node.js, and 12 in dev dependencies. The multi-stage build using distroless drops to 5 CVEs: 2 in the minimal OS layer, 3 in Node.js, and zero in dev dependencies (because they're not included). That's a 92% reduction from a Dockerfile change.



The tradeoff is debuggability. With no shell in the production image, you can't `docker exec` into a running container to poke around. For most production workloads, that's actually a feature – if you need to debug, you should be looking at logs and metrics, not SSH'ing into containers.

Runtime vs Build-Time Scanning

Scanning Strategy

Container scanning can happen at multiple points in the lifecycle. Each stage catches different problems and has different tradeoffs.

Pre-commit scanning Runs against dependency files (package.json, requirements.txt) before code is even committed. Tools like <code>`npm audit`</code> , <code>`pip-audit`</code> , or <code>`trivy fs`</code> catch issues early. This is the fastest feedback loop but only sees declared dependencies, not the full container.	
CI build scanning Runs against the built container image. This is where Trivy shines – you see the complete picture including OS packages, runtime, and application dependencies. This is your primary gate before merge or deploy.	
Registry scanning Runs continuously against images stored in your container registry. Harbor, ECR, and GCR all support this. The key value is catching <code>_new_</code> CVEs in existing images. An image that was clean when you built it might have critical vulnerabilities discovered a week later.	
Runtime scanning Watches running containers for anomalies – processes that shouldn't be running, network connections that shouldn't exist, files being modified that should be immutable. This catches attacks that static scanning can't see, but it's a separate discipline from the build-time scanning we're focused on here.	

Continuous Registry Scanning

New CVEs are published daily. An image that was clean yesterday might have critical vulnerabilities today. Continuous registry scanning catches these without requiring rebuilds.



Most container registries support this natively. Here's how to configure ECR enhanced scanning with Terraform:

ecr-scanning.tf

```

1  # Enable enhanced scanning with Amazon Inspector
2  resource "aws_ecr_registry_scanning_configuration" "main" {
3      scan_type = "ENHANCED"
4
5      rule {
6          scan_frequency = "CONTINUOUS_SCAN"
7          repository_filter {
8              filter      = "*"
9              filter_type = "WILDCARD"
10         }
11     }
12 }
13
14 # Alert on new findings via EventBridge
15 resource "aws_cloudwatch_event_rule" "inspector_findings" {
16     name          = "inspector-vulnerability-findings"
17     description   = "Capture Inspector2 vulnerability findings"
18
19     event_pattern = jsonencode({
20         source      = ["aws.inspector2"]
21         detail-type = ["Inspector2 Finding"]
22     })
23 }
24
25 resource "aws_cloudwatch_event_target" "sns" {
26     rule      = aws_cloudwatch_event_rule.inspector_findings.name
27     target_id = "security-alerts"
28     arn       = aws_sns_topic.security_alerts.arn
29 }

```

AWS ECR enhanced scanning with Terraform.

For Harbor, you can configure Trivy scanning and pull prevention policies through the Harbor Helm chart or API. The key capability is blocking pulls of vulnerable images – if someone tries to deploy an image with critical CVEs, the registry refuses to serve it. This hard stop prevents vulnerable code from reaching production even if CI gates were bypassed.



Handling Common Scenarios

The “Can’t Fix” Situation

Not every CVE (Common Vulnerabilities and Exposures) can be fixed immediately. Here are the scenarios you’ll encounter and how to handle them:

➤ **No patch available:**

The CVE exists, but upstream hasn’t released a fix. Your options are waiting (track the upstream issue), switching to an alternative package, implementing a workaround, or accepting the risk with documentation. CVEs in core libraries like zlib can sit unfixed for months.

➤ **Fix requires breaking changes:**

A patch exists but requires a major version bump. If you’re on React 16 and the fix is in React 18, you’re looking at a significant upgrade project. Options include doing the upgrade, backporting the security fix if it’s simple enough, or accepting the risk temporarily while scheduling the upgrade.

➤ **Transitive dependency:**

The vulnerability is in a dependency of a dependency – something you’ve never directly imported. A CVE in ``nth-check`` that you’ve never heard of, pulled in by ``css-select``, pulled in by ``cheerio``, which you actually use. Your direct dependency might not have updated yet.

For transitive dependencies, package managers offer override mechanisms:

```
1  {
2    "name": "myapp",
3    "overrides": {
4      "nth-check": "2.1.1",
5      "semver": "7.5.4"
6    }
7  }
```

Each ecosystem handles this differently. npm uses `overrides` (yarn uses `resolutions`), Go uses `replace` directives, pip relies on explicit pinning or constraints files, and Ruby lets you declare gems directly even if they’re transitive. Use with caution – you’re overriding what the package author tested against.

With common scenarios handled, the remaining challenge is proving your security posture is actually improving.

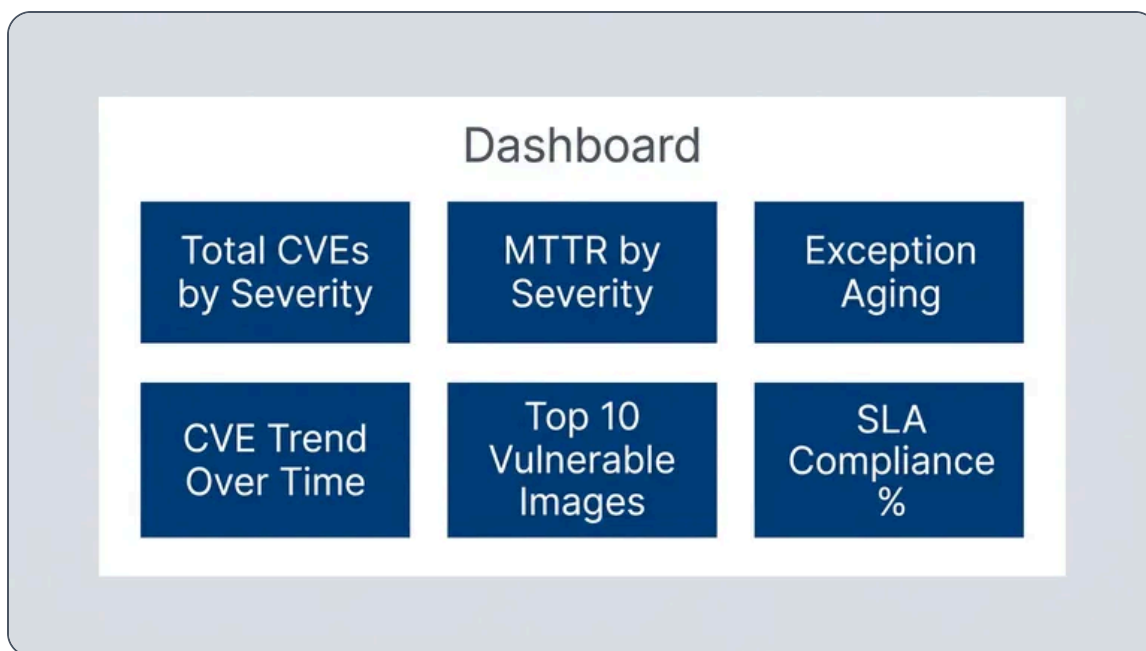


Metrics and Reporting

Security Posture Metrics

Tracking the right metrics tells you whether your security posture is actually improving or just generating noise. Here are the categories that matter:

- ✓ **Point-in-time counts** —show your current state: total vulnerabilities by severity across all images, broken down by fixable vs unfixable. This is your baseline.
- ✓ **Trend metrics** —show direction: mean time to remediate (MTTR) measures the average days from CVE publication to fix deployed. Target less than 14 days for critical vulnerabilities and less than 30 days for high severity. If MTTR is increasing, your process has a bottleneck.
- ✓ **Process metrics** —reveal systemic issues: the number of active vulnerability exceptions should stay manageable. If you have 50+ exceptions, something is wrong — either your policies are too strict, your base images are too old, or teams are gaming the system.
- ✓ **Coverage metrics** —ensure nothing slips through: percentage of deployments with scans should be 100%. Any gap means vulnerable code could reach production undetected.
- ✓ **SLA metrics** —drive accountability: define fix windows by severity (7 days for critical, 30 days for high, 90 days for medium) and track compliance. This turns abstract security goals into measurable commitments.



Security dashboard components.

Reporting to Stakeholders

Different audiences need different views of the same data:

1

Developers need actionable information per PR

What new vulnerabilities did this change introduce, what's fixable, and what commands will remediate them. Deliver this as PR comments or check annotations – immediate feedback in their workflow.

2

Team leads need weekly trends

Are vulnerabilities going up or down, what's the MTTR, how many deployments were blocked. A dashboard or Slack summary works well here.

3

Security teams need daily operational data

New critical findings, pending exception requests, scanner health metrics. They need detailed reports plus alerts for anything requiring immediate attention.

4

Executives need monthly strategic views

Overall posture trend (are we getting better?), SLA compliance percentage, summary of risk exceptions. Keep it high-level – one dashboard, no raw CVE lists.

INFO

The goal of reporting isn't to impress stakeholders with CVE (Common Vulnerabilities and Exposures) counts – it's to drive behavior. Developers need to know what to fix. Executives need to know if the investment in security tooling is paying off.



Integration Patterns

GitHub Security Tab Integration

GitHub's Security tab provides a centralized view of vulnerabilities across your repositories. Trivy can upload findings in SARIF format, which GitHub parses and displays alongside Dependabot alerts and CodeQL results.

.github/workflows/security-scan.yaml

```
1  # Upload findings to GitHub Security tab
2  - name: Upload to GitHub Security
3    uses: github/codeql-action/upload-sarif@v3
4    with:
5      sarif_file: 'trivy-results.sarif'
```

GitHub Security tab integration.

Once uploaded, findings appear in the Security tab under Code scanning alerts. Developers see them in PR checks with inline annotations. The interface is similar to Dependabot – familiar to anyone who's dealt with dependency updates.

Slack Notifications

For teams that live in Slack, immediate notification on scan failures keeps security visible without requiring developers to check dashboards:

.github/workflows/security-scan.yaml

```
1  - name: Notify Slack on Critical CVEs
2    if: failure()
3    uses: slackapi/slack-github-action@v1
4    with:
5      payload: |
6        {
7          "blocks": [
8            {
9              "type": "section",
```




```
10         "text": {
11             "type": "mrkdwn",
12             "text": "🔴 *Security Scan Failed*\n*Repository:* ${github.repository}
13             }\n*Branch:* ${github.ref_name }}"
14         },
15         {
16             "type": "actions",
17             "elements": [
18                 {
19                     "type": "button",
20                     "text": {"type": "plain_text", "text": "View Results"},
21                     "url": "${github.server_url }/${github.repository }/security/code-
22 scanning"
23                 }
24             ]
25         }
26     }
27     env:
28         SLACK_WEBHOOK_URL: ${secrets.SLACK_SECURITY_WEBHOOK }
```

GitHub Action for Slack notification on scan failure.

The `if: failure()` condition means this only fires when the security gate fails – not on every scan. Nobody wants notification fatigue for successful builds.

Conclusion

Shift-left security catches vulnerabilities early when they're cheapest to fix – in the developer's IDE or PR, not in production after a breach. But the goal isn't zero findings; it's actionable findings that developers can actually resolve.

The patterns that make container scanning successful:



Tune policies aggressively.

Report everything, but only fail builds on fixable critical and high vulnerabilities. Unfixable CVEs in the base image shouldn't block feature development.



**Document every exception.**

When you can't fix something, record why in your ignore file with a ticket and expiration date. Undocumented ignores become permanent blind spots.

**Choose minimal base images.**

The single biggest factor in your CVE count is your base image. Distroless or Chainguard images can reduce vulnerabilities by 90% compared to full OS images.

**Generate SBOMs for incident response.**

When the next log4shell drops, you want to answer "are we affected?" in minutes, not days.

**Track trends, not just counts.**

Mean time to remediate and SLA compliance tell you if your security posture is improving. Raw CVE counts just create noise.

Security gates must balance protection with velocity. Gates that block everything get disabled. Gates that developers trust and use – even if they're more permissive – deliver better security outcomes than strict gates that nobody runs.

 SUCCESS

Effective container security isn't about blocking every CVE (Common Vulnerabilities and Exposures) – it's about catching the exploitable ones early, tracking the unfixable ones explicitly, and maintaining developer velocity. A scanner that developers trust and use is infinitely better than a strict scanner that gets disabled.



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<https://www.webstackbuilders.com/articles/container-vulnerability-scanning-ci-shift-left-security>

